

Tab 1

Selection Test: Weather Dashboard

Junior ReactJS Frontend Developer

Create a website that integrates with open weather api (<https://open-meteo.com>), that does the following

As the user lands on the website it displays current weather information. It must fetch gps from browser automatically

Page 1 (Single date)

Current date (or for a given date selected from a calendar view) page

Current date weather parameters (Display individual variables values)

- Temperature (Min, Max, Current)
- Precipitation
- Sunrise time
- Sunset time
- Maximum wind speed
- Relative Humidity
- UV Index
- Precipitation Probability Max
- Air quality
 - PM10
 - PM2.5
 - Carbon Monoxide CO
 - Carbon Dioxide CO₂
 - Nitrogen Dioxide NO₂
 - Sulphur Dioxide SO₂

Hourly data for a selected date

For a given selected date display the parameter in a graph for each hour interval, create individual graphs for each parameter listed below, also chose a suite which graph is suitable for a given data point

Parameters (for each bullet point listed create a individual graph if a more than one parameter is listed in one bullet point then that should be part of one graph)

- Temperature
- Relative Humidity
- Precipitation
- Visibility
- Wind speed (10m)

- Pm10
- pm2.5

Page 2 (2-year data)

This is the second navigation where, when the user lands the user has to select a date range (the maximum range is 2 years) . For the date range selected, the application shall display the values of the data in a chart view (line, bar, etc) a separate chart has to made for each bullet point listed below. The choice of chart type must be taken by observing the data variables.

Date parameters

- **Mean, Max, & Min Temperature**
- **Sunrise & Sunset (Display time as per ist)**
- **Precipitation**
- **Max wind speed & Dominant wind direction**
- **PM10, PM2.5**

Graph Features

- Must scrollable horizontally
- Zoom in and Zoom out

Additional Features

- The website developed must be mobile-friendly, so the components selected to visualize the data must also be populate in mobile view including graphs
- The data must load and render in the website within 500 milliseconds
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Tab 2

Selection Test: Weather Dashboard

Role: Junior ReactJS Frontend Developer

Project Overview

The goal of this test is to develop a responsive web application that integrates with the **Open-Meteo API** (<https://open-meteo.com>) to provide real-time and historical weather insights. The application must automatically detect the user's location via browser GPS upon landing to display localized data.

Technical Requirements

- **Framework:** ReactJS.
- **Data Source:** Open-Meteo API.
- **Performance:** The application must load and render all data within **500 milliseconds**.
- **Responsiveness:** The UI and all data visualizations must be fully mobile-friendly.

Feature Breakdown

Page 1: Current Weather & Hourly Forecast

This page should display information for the current date by default or a specific date selected via a **calendar view**.

1. Individual Weather Variables: Display the following parameters clearly as individual values:

- **Temperature:** Minimum, Maximum, and Current.
- **Atmospheric Conditions:** Precipitation , Relative Humidity , and UV Index.
- **Sun Cycle:** Sunrise and Sunset times.
- **Wind & Air:** Maximum Wind Speed and Precipitation Probability Max.
- **Air Quality Metrics:** Air Quality Index , PM10 , PM2.5 , Carbon Monoxide (CO) , Carbon Dioxide (CO2) , Nitrogen Dioxide (NO2) , and Sulphur Dioxide (SO2).

2. Hourly Data Visualizations: Create individual graphs for each bullet point below, plotted at hourly intervals for the selected date:

- Temperature. (User can toggle from centigrade to fahrenheit)
- Relative Humidity.
- Precipitation.
- Visibility.
- Wind Speed (10m).
- PM10 & PM2.5 (Combined in a single graph).

Page 2: Historical Date range (Max 2 years)

A separate navigation view where users can analyze long-term trends by selecting a date range (up to a maximum of 2 years).

Historical Charts: The developer must choose the most suitable chart type (line, bar, etc.) based on the nature of the variables.

- **Temperature:** Mean, Max, and Min.
- **Sun Cycle:** Sunrise and Sunset (displayed in **IST**).
- **Precipitation:** Total values for the range.
- **Wind:** Max Wind Speed and Dominant Wind Direction.
- **Air Quality:** PM10 and PM2.5 trends.

Visualizations & UI Standards

All graphs and charts must include the following interactive features:

- **Horizontal Scrolling:** For navigating dense datasets.
- **Zoom Functionality:** Ability to zoom in and out of specific timeframes.
- **Mobile Adaptation:** Components must automatically adjust to remain legible on mobile screens.

Submission Instructions

Submit all the items mentioned below through this google form:

<https://forms.gle/hncDckF32CGQkpp69>

1. A **GitHub repository** link with a README.md .
2. Host the website online (plus point)

Evaluation Criteria

1. **Code Quality:** Clean, modular React components.

2. **Performance:** Adherence to the 500ms rendering limit.
3. **Data Handling:** Correct integration with the Open-Meteo API and browser GPS.
4. **UX/UI:** Effective use of charts and responsive design principles.

Timeline

- You have to submit the assignment within **5 days** of receiving it. We expect that the assignment will take you 8 hours to complete, so please schedule your time accordingly . Late submissions will not be accepted.
- You will be informed of the results of your selection test within 3 working days of submitting your assignment, via email.
- If you pass the selection test, you will be invited for the final step – the personal interview. You will have multiple slots to choose from, on Mondays and Thursday. Interviews will last 30-45 minutes

Questions?

Send an email to hr@thelattice.in